vor.GRÜNDEN 2026 · Klagenfurt

CirclOS

The Operating System for Circular & Compliant Manufacturing SMEs

"The platform that turns industrial waste into verified green claims — and revenue."

Green Tech | Circular Economy | Legal Tech / Compliance | AI-Powered

What We'll Cover

01 **The Problem**
Industrial waste + greenwashing compliance crisis

02 **The Solution**
CirclOS — four product layers & the data flywheel

03 **Market Opportunity**
€9.5B waste market + €1.8B compliance SaaS

04 **Product Deep Dive**
AI matching engine + compliance dashboard

05 **Business Model**
Pricing tiers, revenue streams, unit economics

06 **Competitive Landscape**
Why no one can easily copy CirclOS

07 **Technology**
Stack, privacy model, AI approach

08 **Execution Plan**
9-month roadmap week by week

09 **Financials & Funding**
Projections + seed round strategy

10 **Impact & Scale**
Carinthia → EU expansion path

Two Crises — One Platform

Problem 1 — Industrial Waste

- SMEs pay €80–450/tonne for waste disposal
- No intelligent matching between waste producers and potential users
- Existing exchanges are 20-year-old classified-ad boards — no AI
- Waste management has the LOWEST digitalisation of any EU sector
- Materials with real value are destroyed instead of reused

Problem 2 — Greenwashing Crisis

- ECGT Directive 2024/825 — enforcement from Sept 2026
- Bans 'eco-friendly', 'sustainable', 'green', 'climate neutral' without proof
- Nearly all Austrian SMEs use green language in marketing
- Existing SaaS tools (Greenly, Watershed) focus on carbon — NOT claim compliance
- Consultants cost €150–300/hr with no software or evidence generation

KEY INSIGHT: The data generated by solving Problem 1 is exactly the evidence needed to solve Problem 2.

CirclOS — Circular Operating System

The foundational digital layer for circular economy activity and compliance documentation.

L1

Waste Exchange

Free entry tier — list byproduct streams, build the network

→

L2

AI Matching Engine

Paid tier — ML matching, predictive availability, transaction evidence records

→

L3

Compliance Dashboard

Premium SaaS €79–149/mo — ECGT claim scanning + evidence linking

→

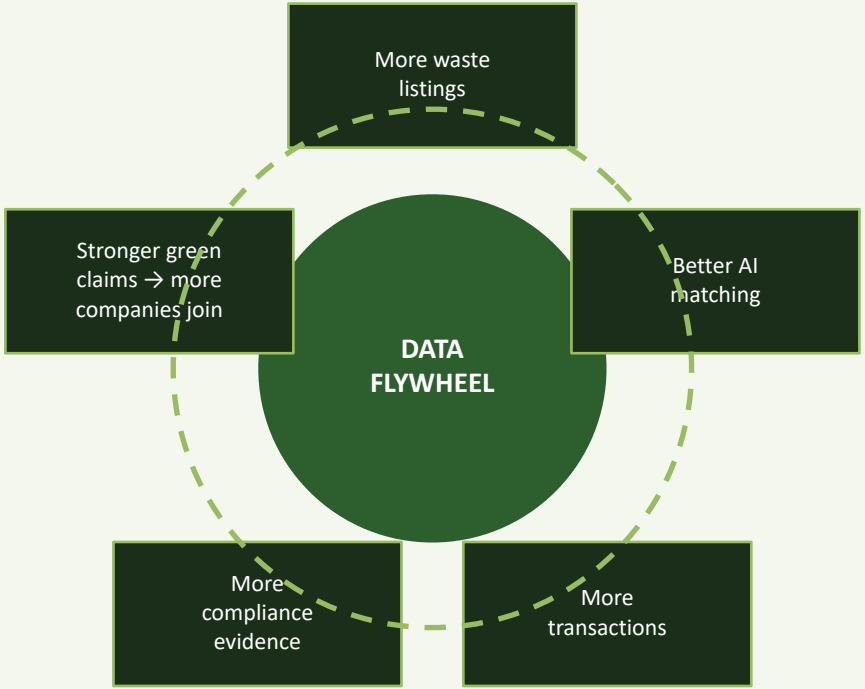
L4

Regulatory Reporting API

Enterprise/B2G — CSRD Scope 3 data, EU taxonomy, government KPIs

ACQUISITION → CORE VALUE → HIGH-MARGIN SaaS → ENTERPRISE

The Industrial Data Network Effect



Supply chain integration effect — not fragile social network effect

Compliance users pressure suppliers to also document → draws more companies in

Each participant makes the platform more valuable for all others

Data moat compounds with every transaction — cannot be bootstrapped by new entrants

A Real Market — Backed by Regulatory Mandates

€9.5B	€1.8B	43,000+	260+
EU Industrial Waste Exchange	EU Green Compliance SaaS Market	Austrian SMEs addressable	Green Tech Valley companies
14% CAGR	Sept 2026 enforcement	via WKO distribution	Year 1 beachhead

Market 1 — Circular Economy Platform

- EU industrial waste exchange: €9.5B · 14% CAGR
- Austrian SME waste disposal cost: €2.8B/year
- CirclOS addressable SME segment: €2.1B
- 20% potential shift from traditional disposal = €560M opportunity

Market 2 — Green Claims Compliance SaaS

- EU SME green compliance tools: €1.8B
- DACH legal compliance SaaS: €4B
- ~280,000 Austrian SMEs using green claims
- €99/month avg ticket = €330M TAM (Year 2–3)

Why Carinthia Is the Perfect Starting Point

Green Tech Valley

260+ companies in Carinthia & Styria already committed to circular economy principles — a ready-made, receptive first market.

Silicon Alps Cluster

Infineon Villach, SAL, Lam Research generate high-value electronics byproducts. Natural platform participants.

Wood K plus & Forestry

St. Veit wood waste research. Carinthian wood processors generate 5–50 tonnes of offcuts monthly needing a match.

KWF Institutional Backing

KWF funding provides government credibility and a warm introduction path to investors and regional authorities.

EU-Compliant Neighbours

Slovenia and northern Italy share identical waste regulations and similar industrial profiles — minimal adaptation for expansion.

University of Klagenfurt

AAU provides AI/ML expertise, academic credibility, and programme support through vor.GRÜNDEN.

How Each Layer Works

Waste Exchange + AI Matching

1. Company creates profile — sector, location, certifications
2. Lists material stream with AWG 2002 legal taxonomy
3. AI matching (3 stages): rule-based compatibility → vector similarity (pgvector) → ML ranking (LightGBM)
4. Both parties notified — CirclOS facilitates introduction
5. Digital sign-off → immutable evidence record created & stored

Compliance Dashboard

1. Web crawler visits every page of company domain
2. German BERT model (gbert-large) identifies environmental claims
3. Each claim classified: ✓ Compliant / ⚠ Needs evidence / ✗ Prohibited (ECGT Art. 6)
4. Evidence checklist generated — exactly what proof is required
5. Claim auto-linked to CirclOS transaction records → legally defensible certificate exported

UNIQUE FEATURE: 'We divert 30% of production waste' → Supported by 14 verified transactions, 8.4 tonnes documented. No competitor can replicate this.

Recurring, Compounding Revenue from Day One

Starter €49/mo	Growth €129/mo	Pro €299/mo	Enterprise Custom
Waste marketplace	AI matching dashboard	Unlimited compliance scans	White-label
Basic listings	Compliance scanner (5 pages)	Monitoring mode	Scope 3 API
Email matching	Monthly certificate	Regulatory exports	Custom reporting
Standard evidence records	+ All Starter	API access	Dedicated support
Small producers	Core SME customer	Mid-size, export-focused	Large cos, gov bodies

Additional Revenue Streams:

- Transaction fees 1–3%
- Certificate renewals
- Data licensing to EU policy bodies
- Enterprise API (CSRD Scope 3)

Conservative Growth — Clear Milestones

Metric	9-Month Programme (Jun 2027)	Year 2 (Dec 2027)	Year 3 (Dec 2028)
Paying customers	**15–25**	**80–120**	**300–500**
Monthly Recurring Revenue	**€1,500–€3,000**	**€10,000–€18,000**	**€40,000–€70,000**
Annual Recurring Revenue	**€18K–€36K**	**€120K–€216K**	**€480K–€840K**
Companies on platform (incl. free)	**30–50**	**200–300**	**1,000+**
Gross Margin (SaaS layers)	**70%**	**75%**	**80%**

FUNDING

Phase 0 (Oct 2026 – Jun 2027): Fully funded by vor.GRÜNDEN + KWF grant (€6,500 consulting budget). No external capital needed.
Post-programme Q3 2027: Target €300K–€500K pre-seed from aws Gründerfonds · Ananda Impact Ventures · Green Generation Fund.

Where the Gaps Are — and Why CirclOS Wins

Competitor	Type	Key Gap vs CirclOS	Threat
Recy.net / Recycling Börsen	Waste Exchange	Manual listings, no AI, no compliance layer, 20yr-old model	Low
Scraphero	Waste Exchange	Metal-specific only, no compliance, no Austrian focus	Low
Greenly / Watershed	Carbon SaaS	Carbon footprint only, English, enterprise pricing €500+/mo	Medium
EcoVadis	ESG Rating	Annual audit model, no marketing claim scanning, no marketplace	Low
SAP Sustainability	Enterprise ERP	€50K+ implementation, not accessible to SMEs	Low (short-term)
Austrian Consultants	Prof. Services	€150–300/hr, no software, no evidence generation	Substitute

OUR MOAT

Data network effect (transaction history) · Compliance evidence linkage (only possible with marketplace data) · Austrian legal calibration (6-month NLP investment) · Green Tech Valley / WKO relationships (years to build, cannot be purchased) · First-mover advantage: every month adds data competitors must replicate from zero.

Built to Scale — Designed for Privacy

Backend API	**Python 3.12 + FastAPI** Async performance, auto-generated OpenAPI docs, large DACH developer pool	**NLP Compliance**	**deepset/gbert-large (fine-tuned)** Best German BERT model, fine-tuned on Austrian green marketing corpus
Frontend	**React 18 + TypeScript** Component reusability for complex dashboards, Vite for fast dev	**Hosting**	**AWS eu-central-1 (Frankfurt)** GDPR-compliant EU data residency — required by enterprise customers
Database	**PostgreSQL 16 + pgvector** Semantic similarity without separate vector DB, ACID compliance for records	**Auth**	**Supabase Auth (MVP)** Rapid implementation; custom OIDC post-Series A
AI Matching	**sentence-transformers + LightGBM** Material encoding + gradient boosting ranker (interpretable, works on small datasets)	**Payments**	**Stripe + Austrian VAT** Handles Austrian 20% VAT, B2B reverse charge, subscription management

Oct 2026 – Jun 2027: Four Phases, Non-Negotiable Order

Phase 1

Weeks 1–4
Oct 2026

Foundation

- Legal waste taxonomy audit (AWG 2002)
- Data model & privacy architecture design
- Tech stack commitment & team roles
- Green Tech Valley — 5+ meetings booked

✓ Legal memo + ER diagram + 5 meetings

Phase 2

Weeks 5–9
Nov 2026

Anchor Partners

- Waste audits with 5–8 Carinthian companies
- Pricing sensitivity testing
- 3 founding member LOIs signed
- Manual compliance scans → urgency created

✓ 3 signed LOIs + 8 audit reports

Phase 3

Wks 10–18
Dec–Feb 2027

Waste Exchange MVP

- Profiles, listings, matching UI built
- First 3 live transactions facilitated
- Evidence record PDF validated
- 10+ companies onboarded, ML ranking added

✓ Live platform + 3 transactions + traction metrics

Phase 4

Wks 19–36
Mar–Jun 2027

Compliance Layer

- Website scanner + BERT claim classifier
- Compliance dashboard + evidence linking
- Pricing launch + founding members to paid
- WKO partnership + Demo Day preparation

✓ First paying customers + MRR + seed deck

The Legal Landscape That Creates the Market

BINDING — enforcement Sept 2026

EU ECGT Directive 2024/825

Bans generic unsubstantiated environmental claims: 'eco-friendly', 'green', 'sustainable', 'climate neutral' (unless verified lifecycle reductions, not offsets). Austria must transpose by March 2026. Enforcement window is closing.

NATIONAL LAW — governs all waste handling

Austrian AWG 2002

Distinguishes legally between 'Abfall' (waste) and 'Sekundärrohstoff' (secondary raw material). Critical for CirclOS marketplace: secondary raw materials trade freely; waste requires licensed facilities. CirclOS onboarding enforces this classification.

EU LAW — hierarchy & end-of-waste criteria

EU Waste Framework Directive 2008/98/EC

Establishes waste hierarchy (prevention → reuse → recycling → recovery → disposal). Defines 'end-of-waste' criteria. CirclOS transactions contribute measurably to EU progress on this hierarchy — reportable for CSRD.

WITHDRAWN Jun 2025 — re-proposal uncertain

EU Green Claims Directive (proposed)

Would have required pre-approval of claims by national authorities. Withdrawn due to SME burden concerns. However, ECGT remains in full force. German/Austrian courts already enforcing independently. Compliance market exists regardless of GCD status.

Honest Assessment — and How We Manage It

CRITICAL	**Two-sided marketplace cold start** MITIGATION: Secure binding LOIs from min. 3 sellers AND 3 buyers before launch. Partner with Green Tech Valley to digitise their existing manual industrial symbiosis matchmaking.
CRITICAL	**Regulatory stagnation on green claims** MITIGATION: Build waste exchange as the primary product — compliance is the upsell. If compliance urgency fades, marketplace still stands. German/Austrian courts enforce independently anyway.
SIGNIFICANT	**Legal liability from compliance tool** MITIGATION: Explicit T&C: tool provides guidance, not legal advice. Co-sign methodology with Austrian law firm. Frame as demonstrating good-faith effort. Mandatory disclaimer on all certificates.
SIGNIFICANT	**SME data reluctance** MITIGATION: Privacy-by-design: only material category and volume ranges visible to others. Exact schedules are private. Present the privacy model upfront — it's a selling point.
MODERATE	**Large player copies the concept** MITIGATION: Moat is data and relationships, not software. A company with 2 years of CirclOS transaction history will not switch. Get deep before wide. First-mover depth is the defence.

Carinthia → EU: Step by Step

1 Carinthia
Oct 2026 – Jun 2027

Green Tech Valley (260+), Wood K plus, Silicon Alps. Target: 50 companies, 15–25 paying compliance customers.

2 Styria
Q3 2027

Natural extension — same Green Tech Valley cluster. Graz industrial & TU Graz ecosystem. 1 additional BD hire in Graz.

3 Austria-wide
Q4 2027

WKO national partnership. Upper Austria (Linz heavy industry), Vorarlberg (textiles/plastics). No product changes needed.

4 Bavaria / Germany
Q1 2028

Same language, strictest courts in EU on green claims. Munich + Nuremberg clusters. Adjust compliance for UWG. 10× Austria market.

5 Switzerland
Q2 2028

German-speaking, high SME density, strong sustainability culture. Localise compliance tool for Swiss OR.

6 EU-wide Platform
2028–2029

License to industrial clusters (France/SUEZ, Poland, Netherlands). Regulatory Reporting API becomes the key product for cross-border flow data.

Every Transaction Is an Environmental and Economic Event

Environmental Impact

- Direct diversion of material from landfill or incineration
- Reduces virgin material demand → lower extraction footprint
- Local matching prioritised → reduces industrial transport emissions
- Measurable contribution to EU Circular Economy Action Plan KPIs

Economic Impact

- Converts disposal costs (€80–450/tonne) into revenue or savings
- Reduces raw material procurement costs for receiving companies
- Creates economic value from materials previously destroyed
- SME compliance tool prevents fines of €5K–€50K range

Social / Governance Impact

- Democratises legal compliance: €99/month vs €300/hour consultant
- Levels playing field between large corps (with compliance teams) and SMEs
- Creates supply chain transparency — stakeholders verify sustainability claims
- Transaction data shared anonymised with AAU for policy research

Directly aligned with UN SDG 12 (Responsible Consumption & Production) · SDG 13 (Climate Action) · Austrian National Circular Economy Strategy

Team Structure & University Anchoring

Technical Co-Founder

- Python + FastAPI backend development
- ML/NLP: fine-tuning BERT models, LightGBM rankers
- Database design (PostgreSQL + pgvector)
- React frontend for dashboard UIs
- AWS infrastructure and deployment

Commercial Co-Founder

- B2B sales and partnership development
- Waste audit fieldwork and company interviews
- Regulatory research (Austrian environmental & marketing law)
- WKO and Green Tech Valley relationship management
- Investor relations and programme reporting

AAU Anchor

Ideal host: AAU Department of Economics/QED (financial & regulatory modelling) + Institute of Applied Informatics (AI/ML supervision).
Budget allocation: €2,500 environmental law firm · €2,000 AI/ML specialist · €1,500 UX designer · €500 events & networking.

Why CirclOS, Why Now, Why Here

CirclOS

The Operating System for
Circular & Compliant Manufacturing SMEs

- September 2026 ECGT deadline creates acute, time-bound compliance demand
- Carinthia's Green Tech Valley — 260+ ready companies, institutional backing
- Two-product flywheel: each layer makes the other more valuable
- No existing tool combines AI waste matching with verified compliance evidence
- Every month of operation adds data competitors must replicate from zero